

DAI 1000

Digital and AI Literacy

L6 - Machine Learning Essential II

Objectives

- Understand key AI paradigms and foundational concepts that underpin modern intelligent systems
- Explore deep learning principles and architectures that enable advanced perception and decision-making
- Appreciate the applications of different AI algorithms across diverse real-world domains and disciplines

Content

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Deep Learning

Paradigm →
Tasks →
Algorithms/ Models →

Supervised Learning

Classification

Regression

KNN, linear models, DNNs ...

Semi/Unsupervised Learning

Clustering

*Semi-sup
Learning*

*Generative
Model*

....

K-means, PCA, DNNs ...

Reinforcement Learning

Machine Learning Paradigm

ML is a broad field commonly divided into three main paradigms, with different tasks and algorithms for each approach.

Supervised Learning

The model learns from
labeled data

Unsupervised Learning

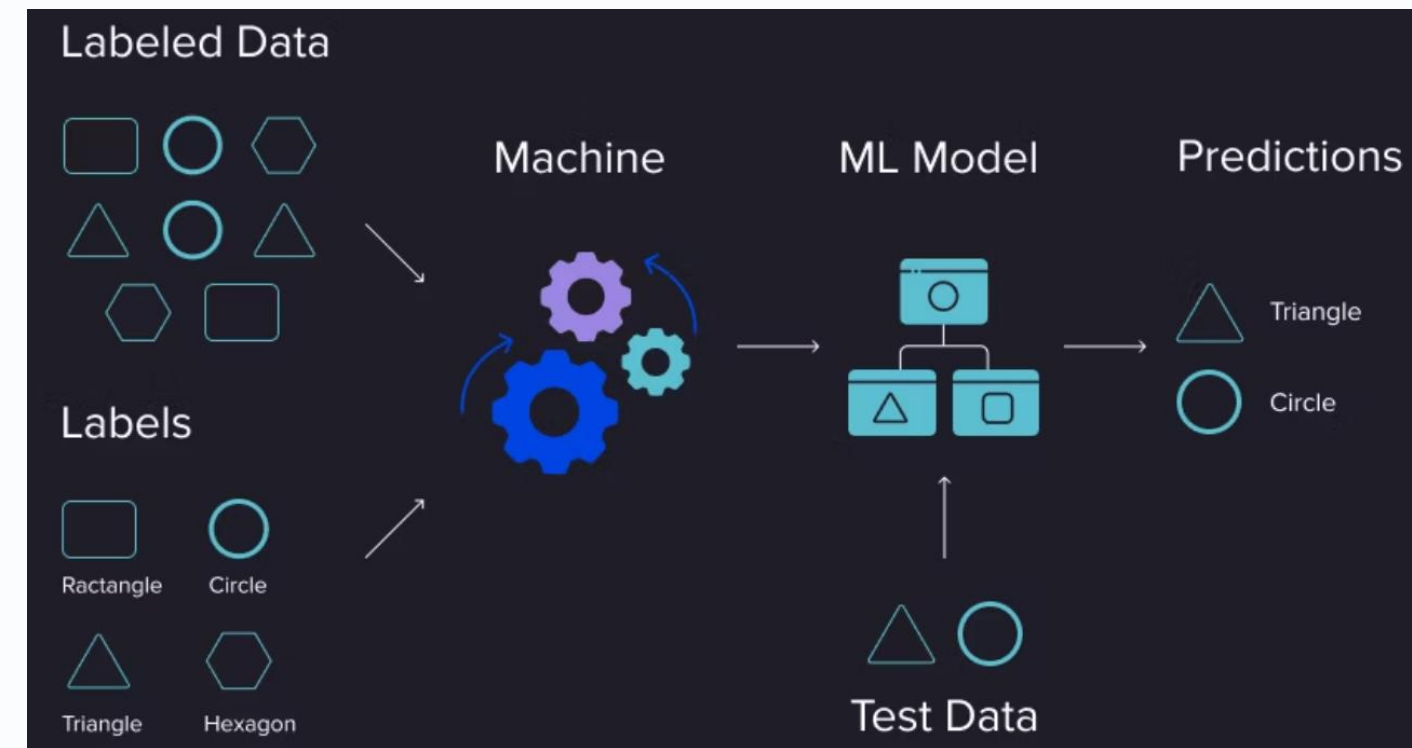
The model uncovers
patterns in **unlabeled**
data

Reinforcement Learning

The agent learns via
feedback from the
environment

Supervised Learning?

- Use labeled training datasets to understand relationships between inputs and outputs
- Once trained, the model can predict correct outputs based on new, unlabeled real-world input data

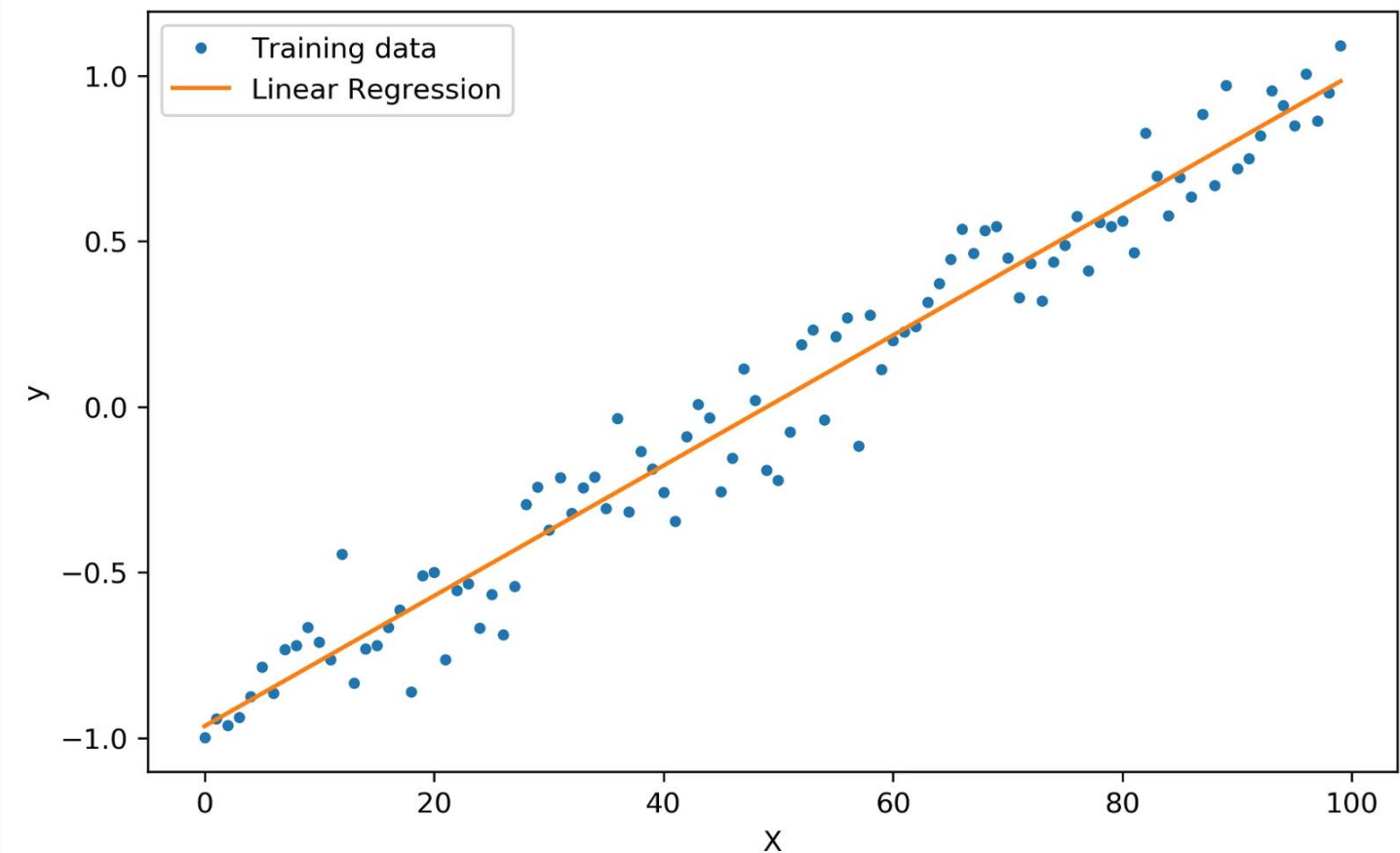


Main tasks: Classification and Regression

Supervised Learning – Regression

In regression problems, the dependent variable is a **continuous value**, and models attempt to predict the target output.

Examples of regression algorithms:
Linear regression, logistical regression
and polynomial regression.



Regression Examples

House Price Prediction

To estimate property values based on property's details

Input: Room Number, square footage, location, year built, nearby schools

Output: Predicted house price

Medical Risk Prediction

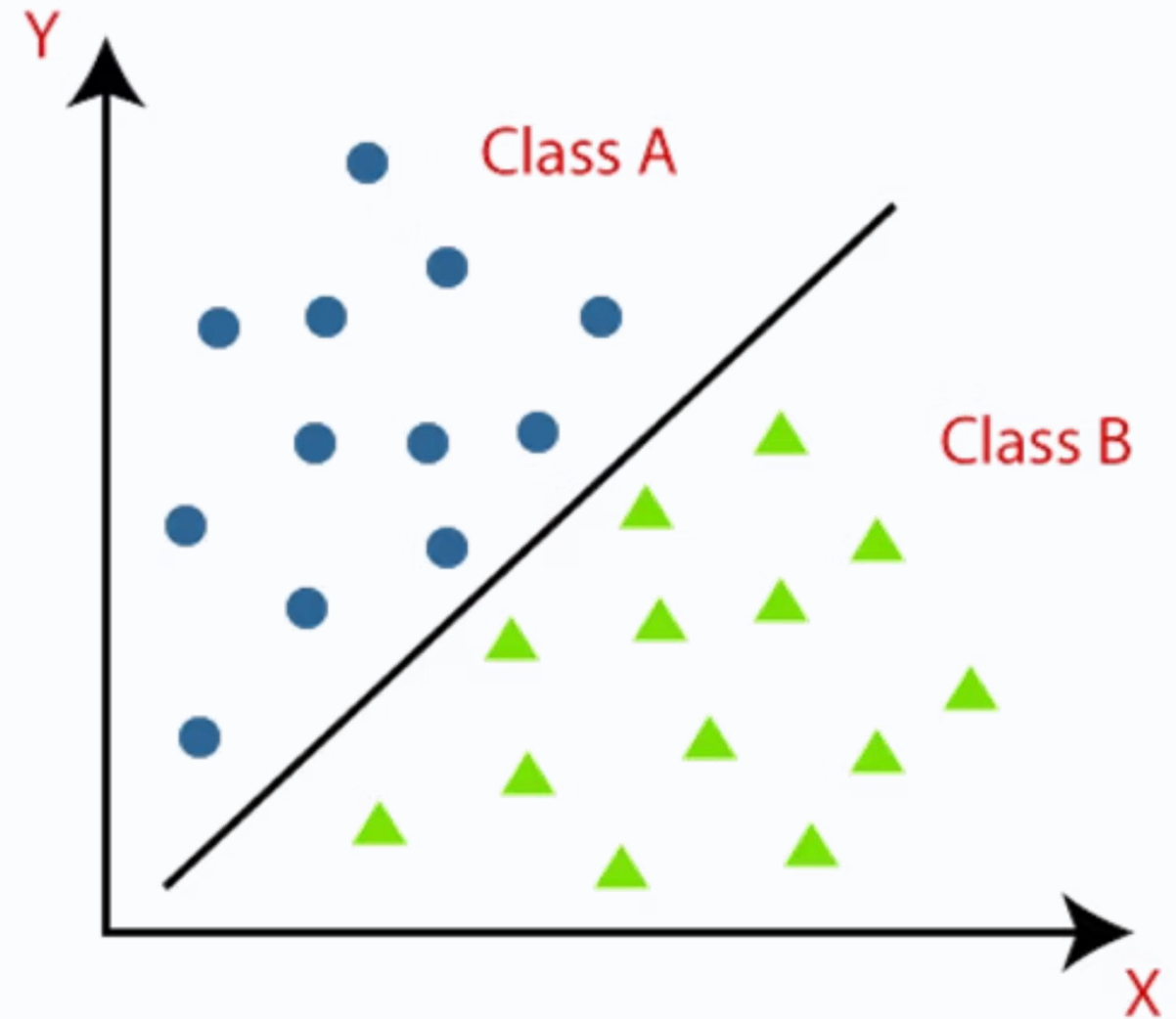
Healthcare providers assess patient risk and personalize treatment

Input: Age, weight, medical history, lifestyle factors

Output: Predicted blood pressure

Supervised Learning - Classification

It uses algorithms to **sort data into two or more categories**, recognizing specific entities and determining how they should be labeled.



Classification Examples

Binary Classification

Distinguishes between two classes

Email Spam Detection

Input: Email subject, body text, sender address, suspicious keywords

Output: Spam or Not Spam

Use case: Gmail and Outlook filter unwanted emails

Classification Examples

Multi-class Classification

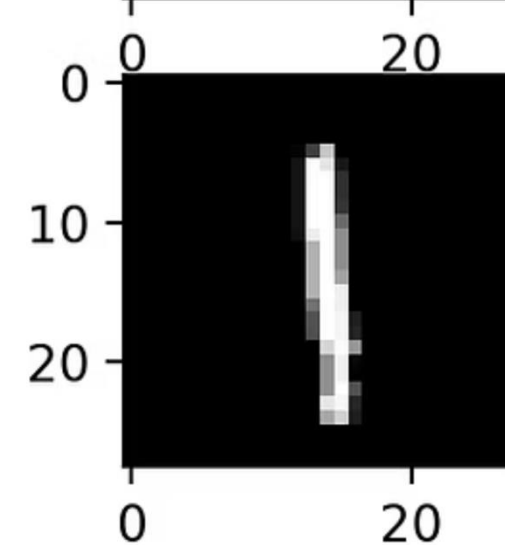
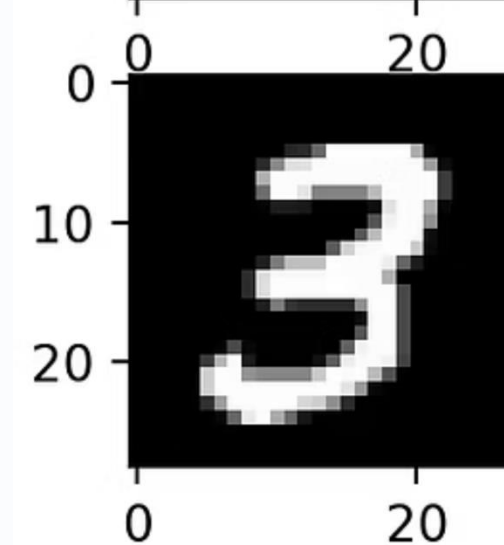
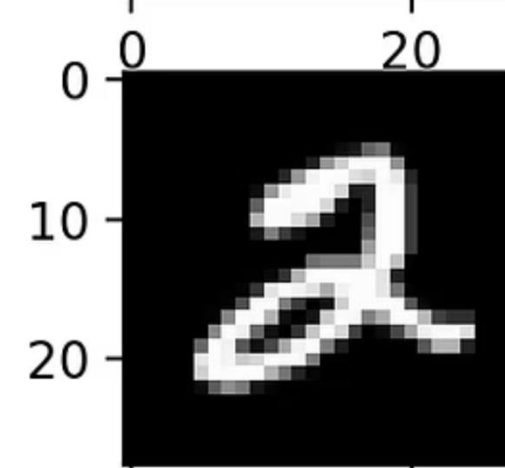
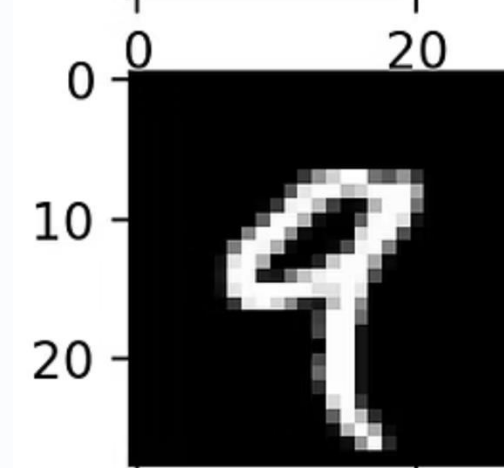
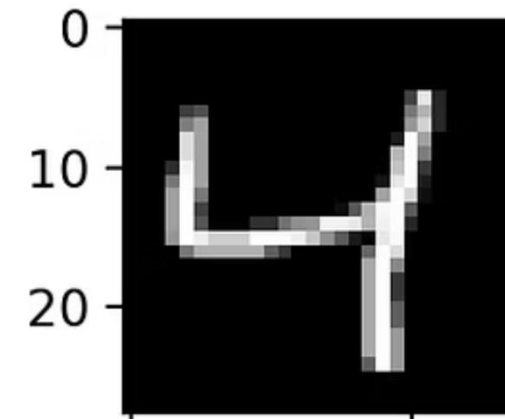
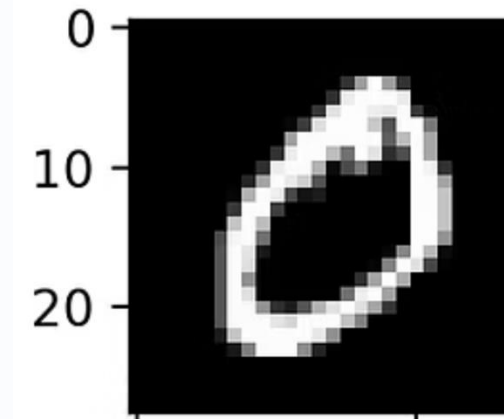
Categorizes data into multiple classes

Handwriting Recognition

Input: Pixel values of handwritten digits

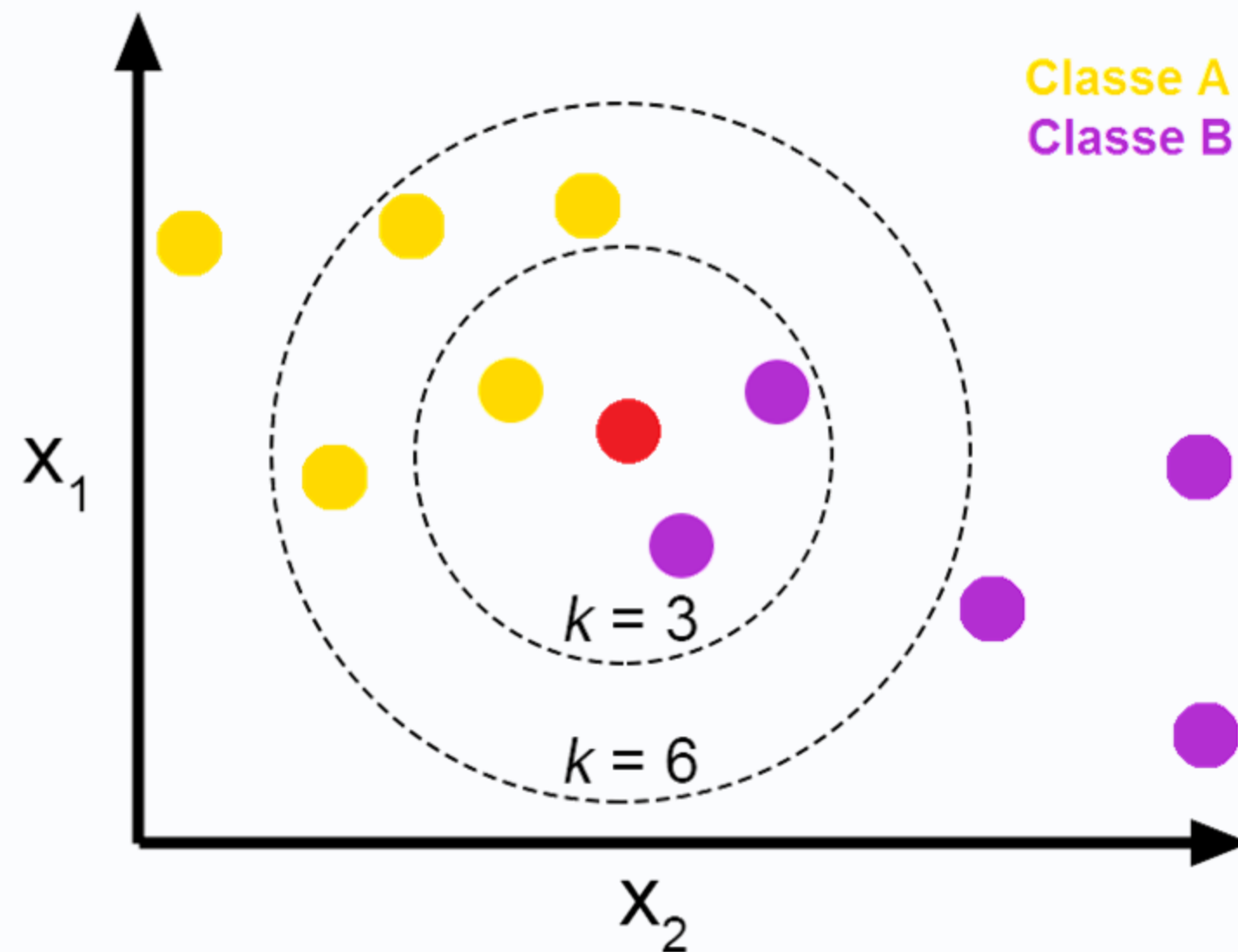
Output: Digits 0–9

Use case: Postal mail processing, bank check processing, form data entry



K-nearest neighbors

<https://www.bilibili.com/video/BV1iU49zYEaB/>



Unsupervised Learning

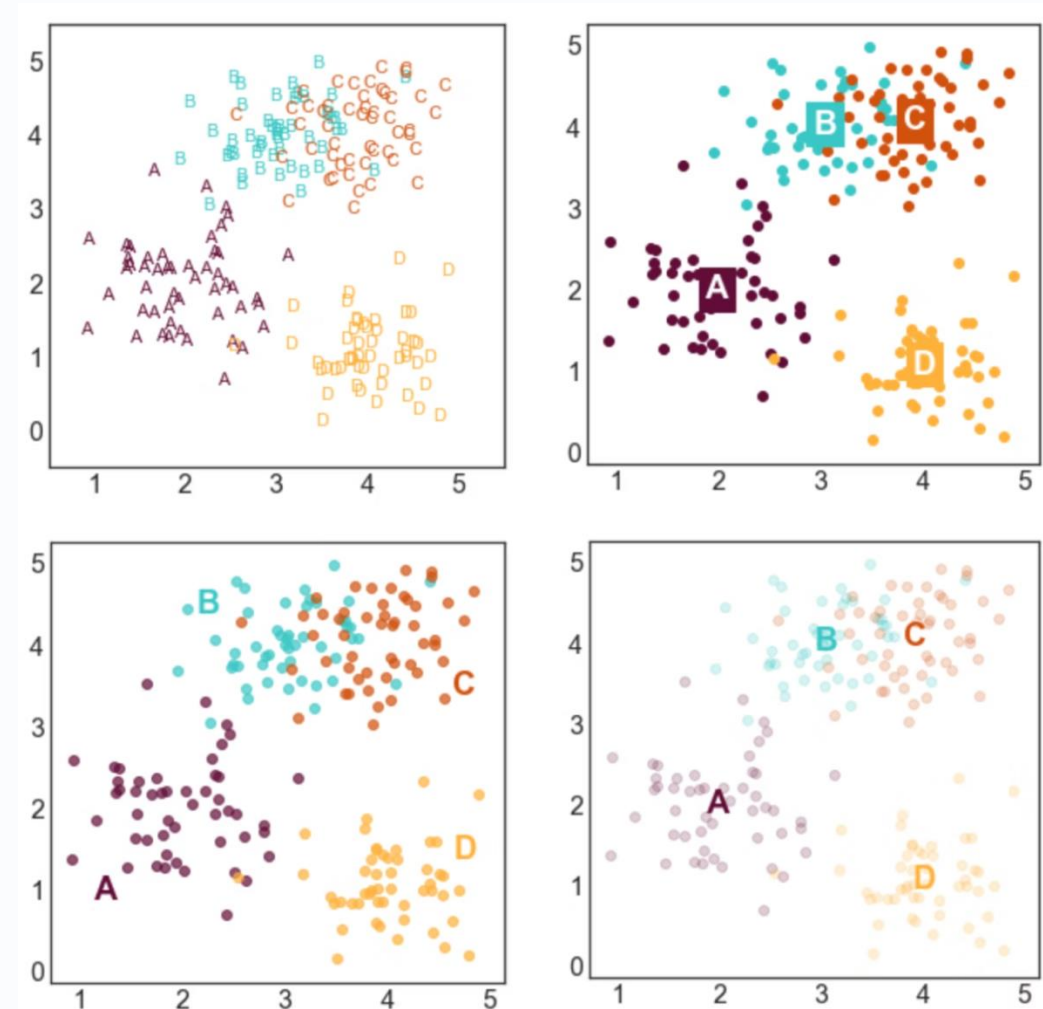
It can analyze and cluster unlabeled data sets, discovering hidden patterns or data groupings without the need for human intervention.

Unsupervised learning's ability to discover similarities and differences in information makes it the ideal solution for exploratory data analysis.

Tasks: Clustering, Association Rules, and etc.

Unsupervised Learning –Clustering

Clustering is used to process raw, unclassified data objects into groups represented by structures or patterns in the information.



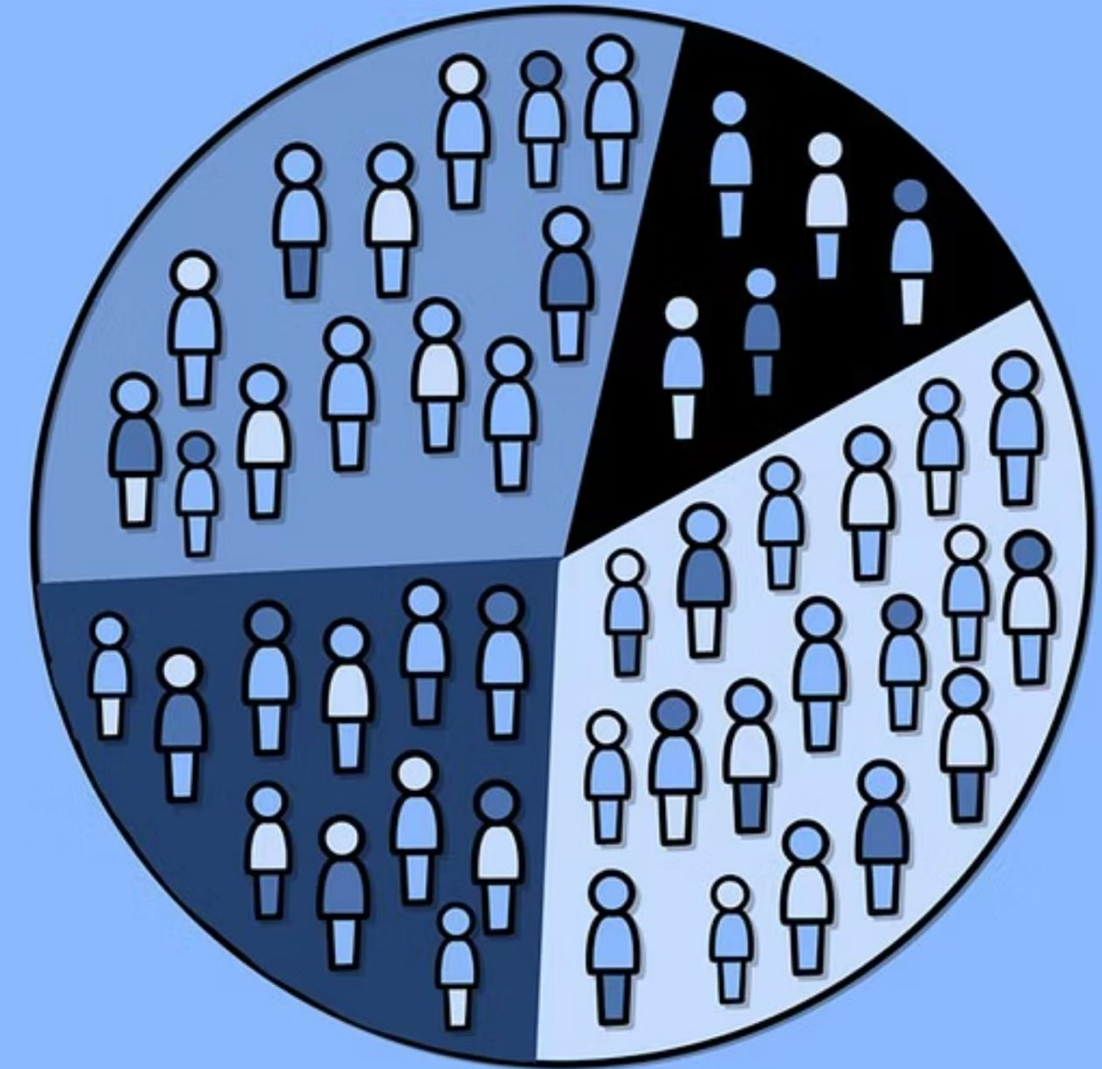
Clustering Examples

Customer Segmentation in Marketing

Input data: purchase history, browsing behavior, demographics, spending habits

Output: groups of customers with similar behavior (e.g., "bargain hunters," "loyal high-spenders")

Use case: Businesses use clustering to tailor marketing strategies, personalize offers, and design loyalty programs.



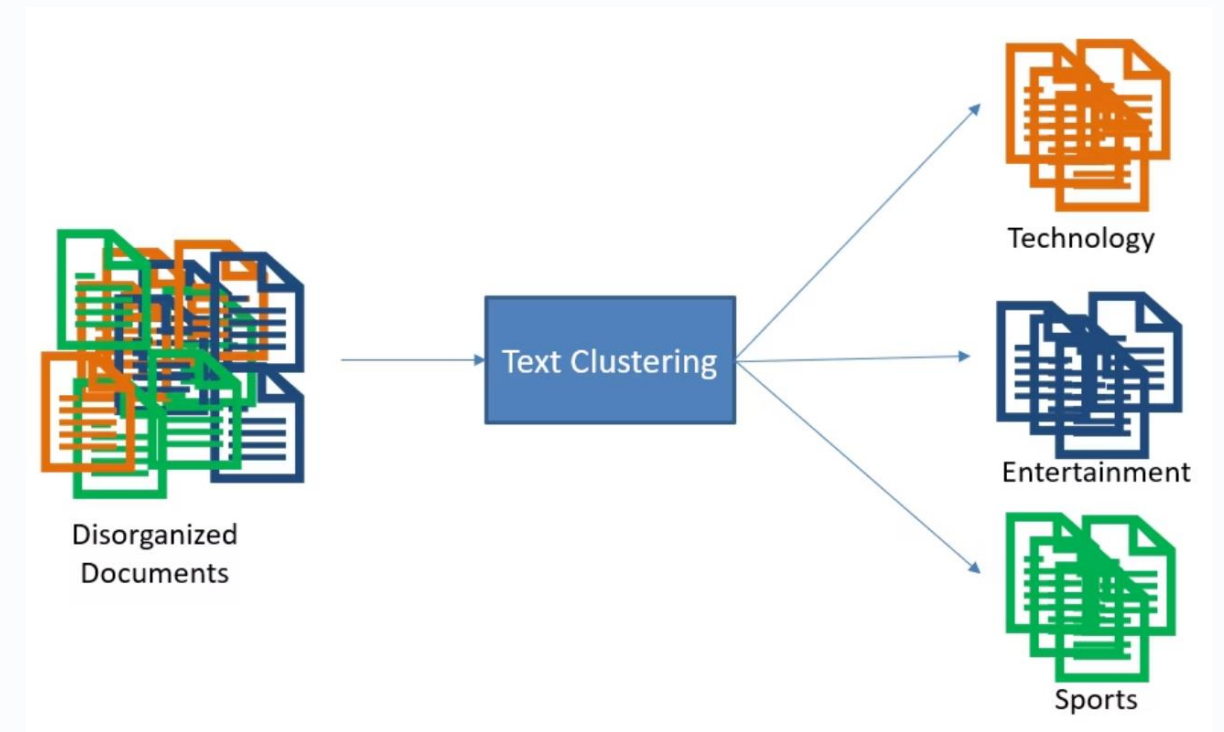
Clustering Examples

Document or News Article Grouping

News aggregators and search engines automatically group content to help users navigate large corpora

Input data: content or keywords from articles, web pages, or reports

Output: clusters of documents on similar topics (e.g., politics, sports, health)



K-Means Clustering Demo

<https://www.naftaliharris.com/blog/visualizing-k-means-clustering/>

Unsupervised Learning – Association Rules

An association rule is a rule-based method for finding relationships between variables in a given dataset.

Market Basket Analysis

Retailers optimize product placement and cross-selling strategies.

Input: customer transaction records

Identified Rule: If a customer buys bread and butter, they're likely to buy jam.

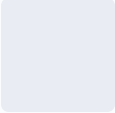
Online Recommendation Systems

Netflix, YouTube, and Amazon recommend content based on user behavior.

Input: user interaction logs

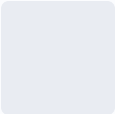
Identified Rule: Users who watched Movie A and B tend to watch Movie C.

Limitations of Unsupervised Learning



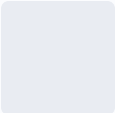
Computational Complexity

High volume of unlabeled training data creates processing challenges



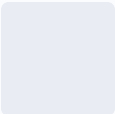
Longer Training Times

Extended processing periods required for analysis



Higher Risk of Inaccurate Results

Without labeled data, results may become inaccurate



Lack of Transparency

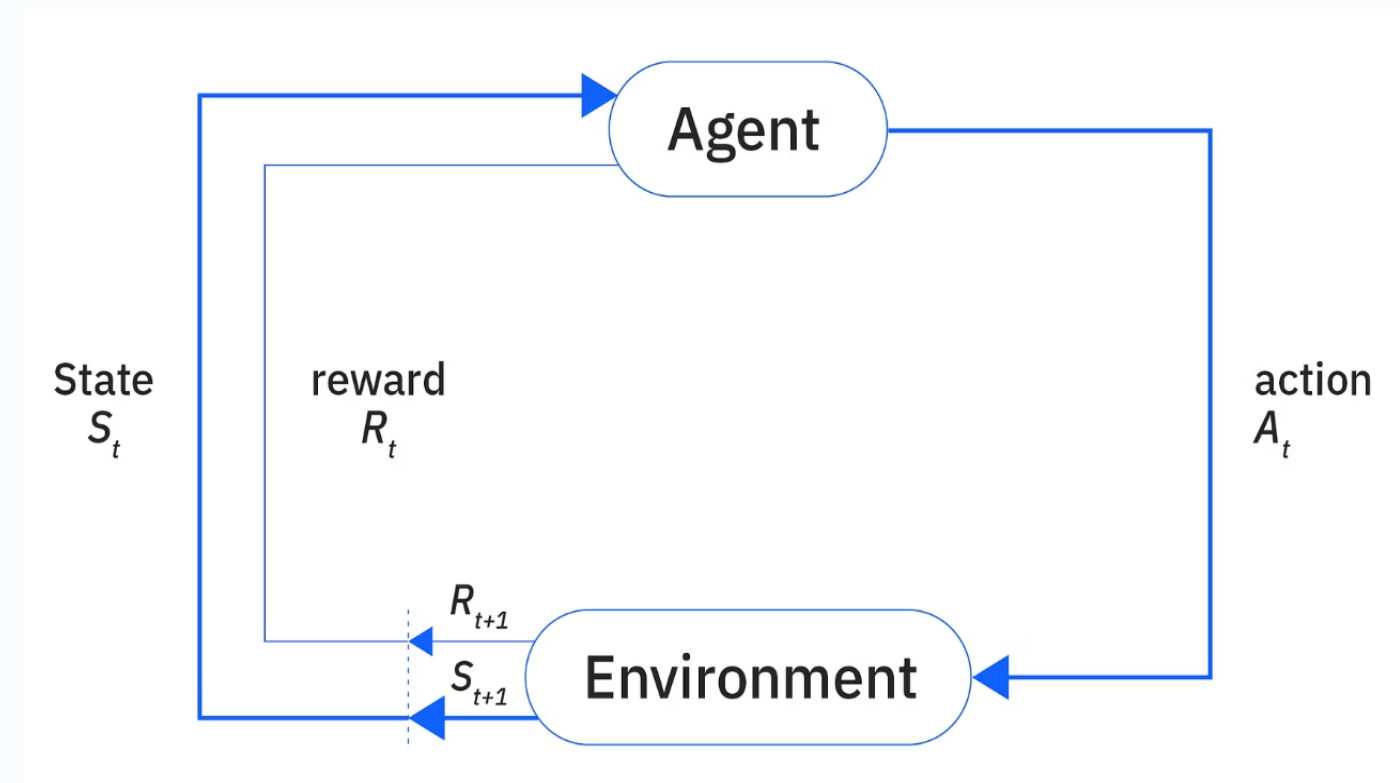
Unclear basis on which data was clustered

Reinforcement Learning

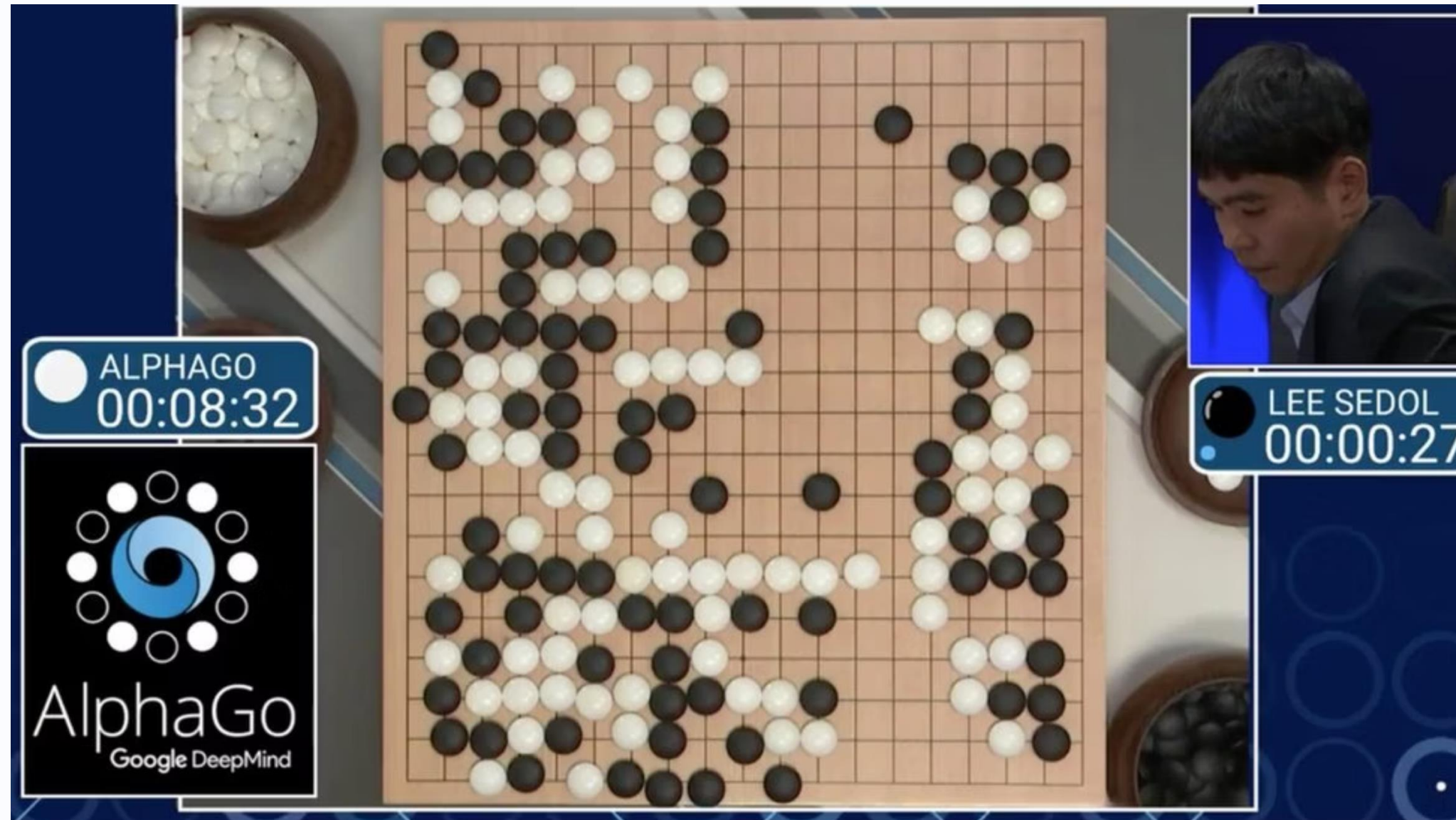
Reinforcement learning is concerned with how an autonomous agent should take actions in a dynamic environment in order to **maximize a reward**

In reinforcement learning, an autonomous agent learns to perform a task by **trial and error** in the absence of any guidance

Examples: Robots and self-driving cars



Reinforcement Learning Examples



Reinforcement Learning Examples

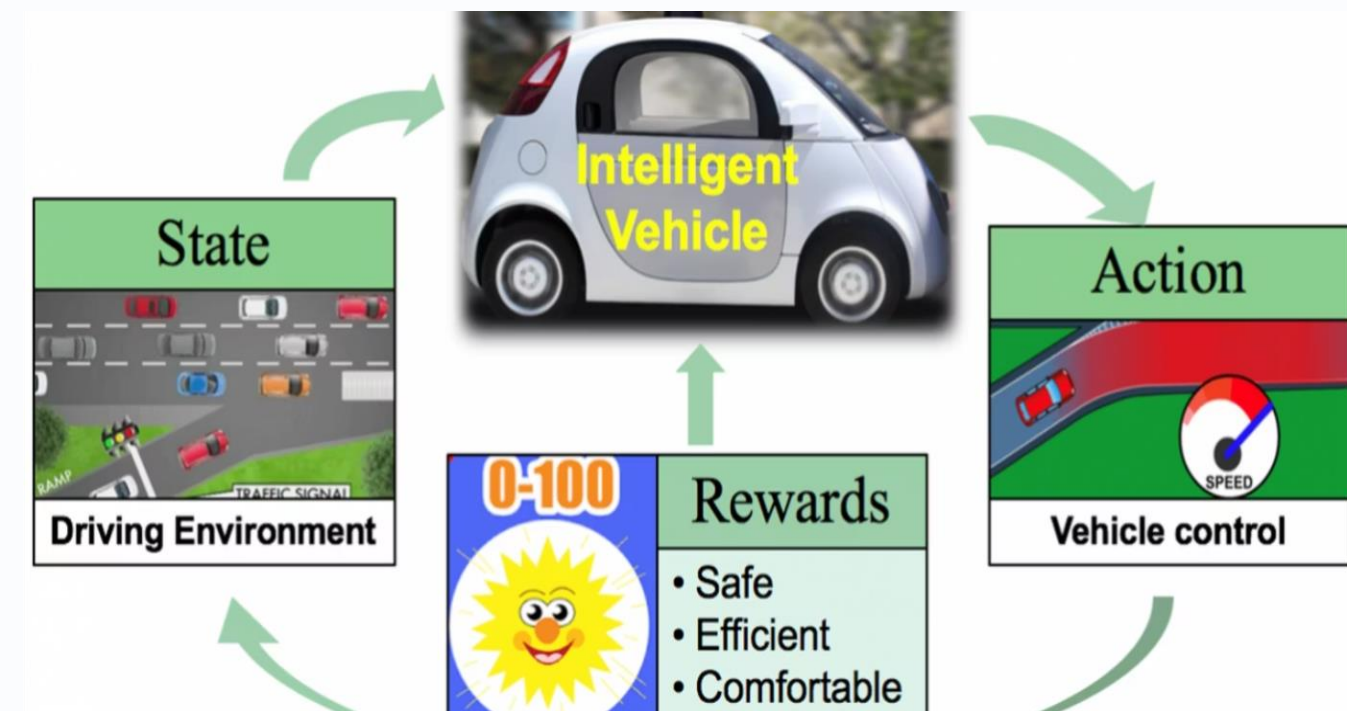
Self-Driving Cars

Agent: Self-driving car

Actions: Accelerate, brake, turn, yield

Environment: Road traffic, pedestrians, traffic lights

Reward: Safe and efficient driving (staying in lane, avoiding collisions)



Comparison

Type	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Input Data	Labeled	Unlabeled	Feedback from environment
Goal	Predict output for known input	Discover structure in data	Maximize reward through action
Examples	Spam detection, cat/dog classification	Customer segmentation, anomaly detection	Self-driving cars, ad personalization
Limitations	Needs labels, prone to overfitting	Hard to interpret, high complexity	Slow training, Specifying reward difficulty, and Generalization

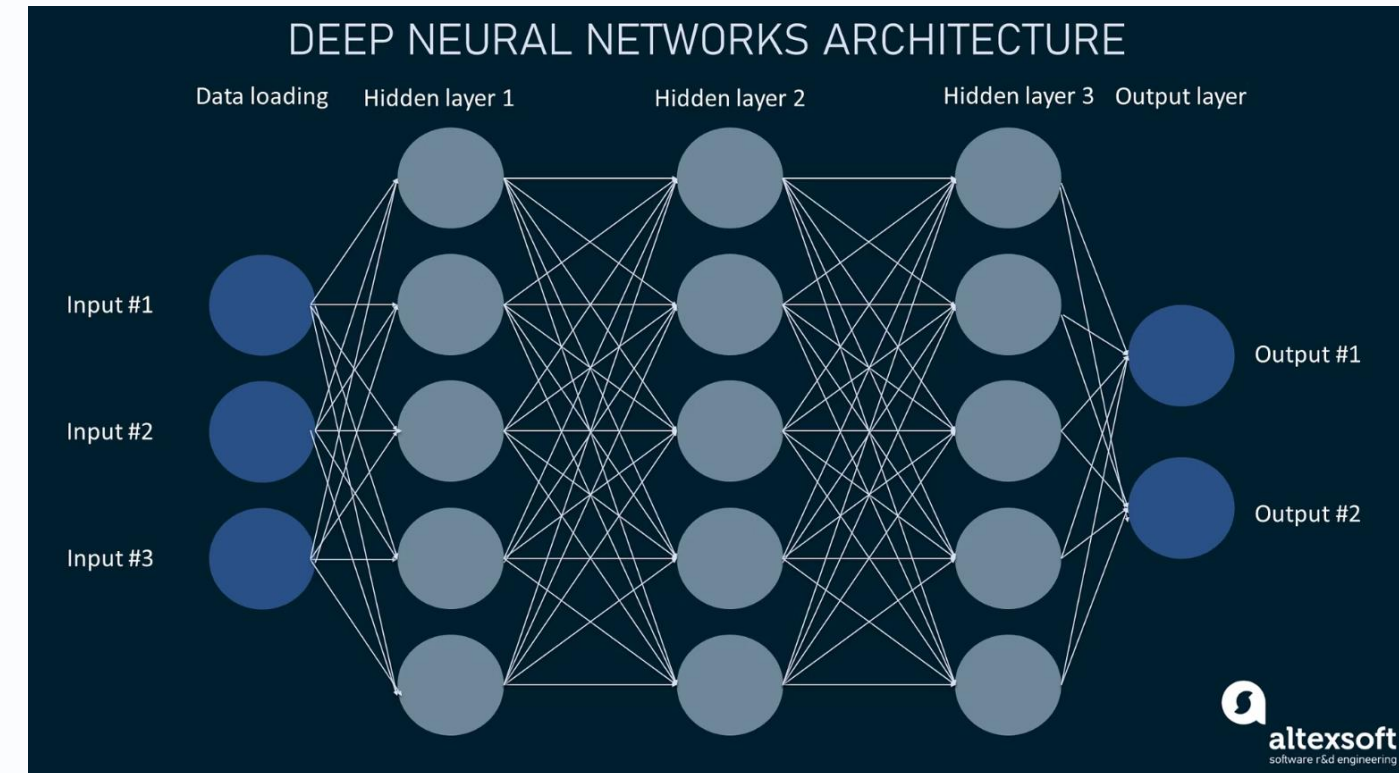
Content

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Deep Learning

What is Deep Learning?

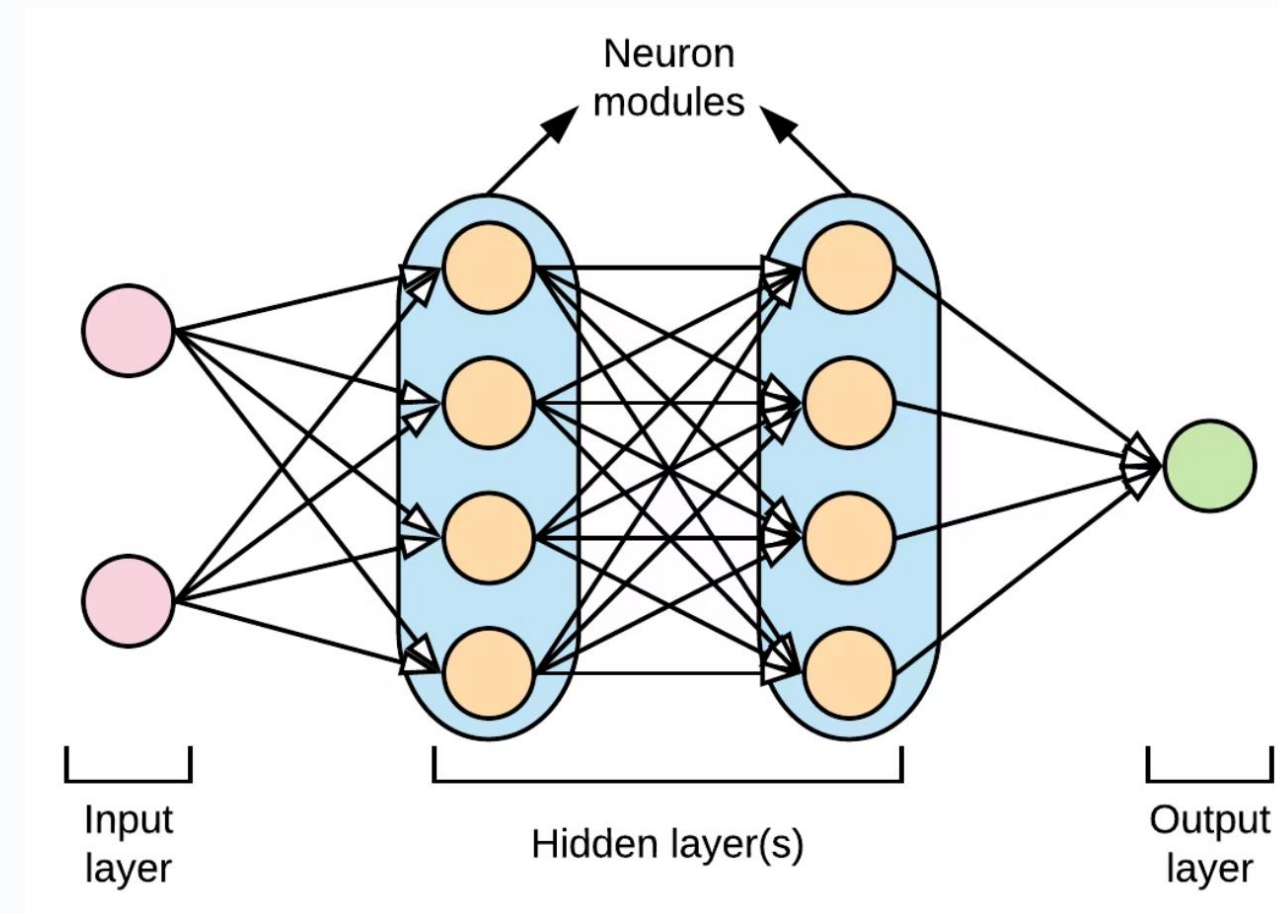
A subset of machine learning

Uses multilayered neural networks, called **deep neural networks**, to make predictions



What is Deep Learning?

- **Input layer:** holds the features ($X_1, X_2, X_3, \dots$)
- **Hidden layers:** consist of artificial neurons (or nodes) that transform inputs into new representations
- **Output layer:** produce the final prediction (such as a number for regression, or a probability distribution for classification)





Neural Networks Explained in 5 minu

Learn more about watsonx: <https://ibm.biz/BdvxRs> Neural networks reflect the behavior of the human brain, allowing

<https://youtu.be/jmmW0F0biz0>

How Does a Neural Network Work?

Should I go party tonight?
(1→GO, 0→Don't GO)

$$\hat{y} = \sum w_i x_i + \text{bias} = w_1 x_1 + w_2 x_2 + w_3 x_3 + \text{bias}$$



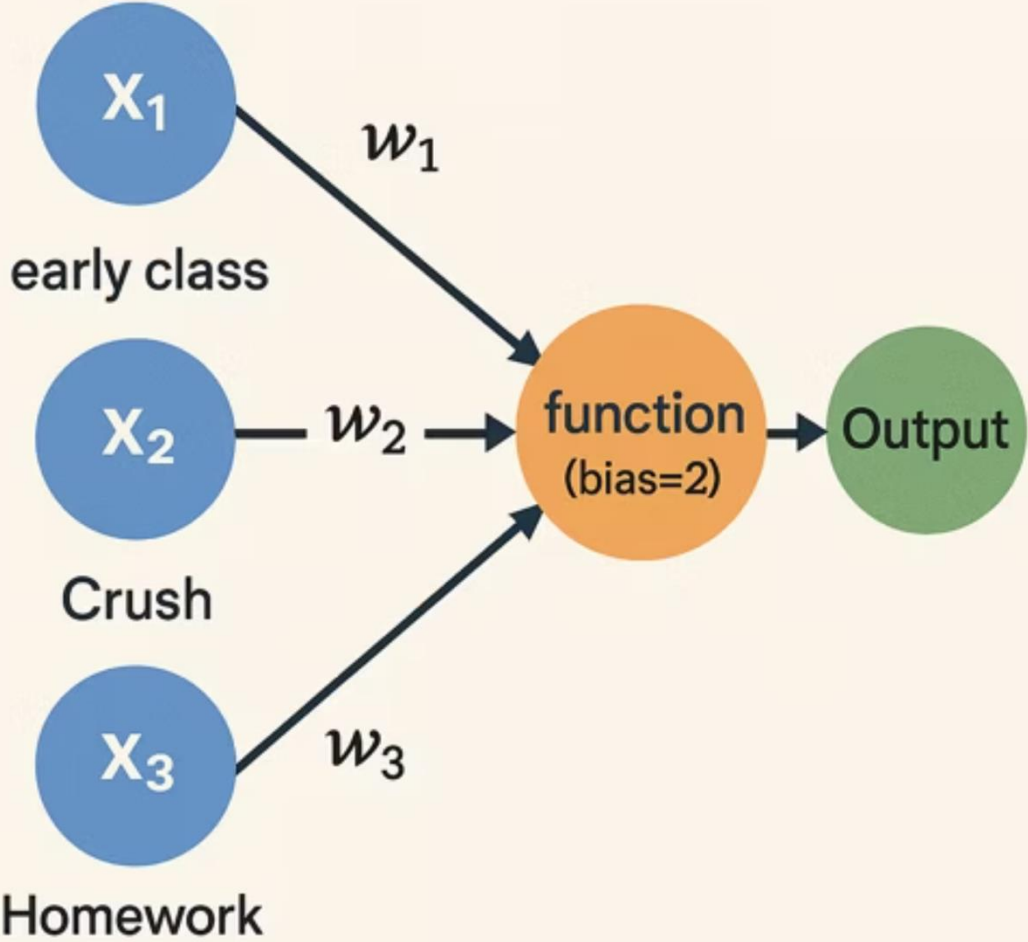
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This result passes through an activation function that makes a binary decision:

If	Output
$\sum w_i x_i + b \geq 0$	$f(x) = 1$
$\sum w_i x_i + b < 0$	$f(x) = 0$



Should You Go to the Party Tonight?

The predicted outcome ($\hat{y}$) depends on three binary factors:

 Early Class?

Yes: 1, No: 0

 Will Crush Be There?

Yes: 1, No: 0

 Homework Finished?

Yes: 0, No: 1



Assigning Weights and Bias



Weights (w)

Each factor gets a weight representing its importance in your decision:

- $w1 = -6$: You really care about early classes
- $w2 = 5$: Seeing your crush is a big motivator
- $w3 = -3$: You're concerned about unfinished homework

Bias

This represents your baseline inclination before considering any specific factors

You have a natural tendency to go out, expressed as a bias of **+2**

Running the Calculation

✓ Input Values (x)

- **x1 = 0**: No early class tomorrow
- **x2 = 1**: Your crush will be at the party
- **x3 = 1**: You haven't finished your homework

1234 Final Calculation

Plugging into the formula:

$$\begin{aligned} \hat{y} &= (0 \times -6) + (1 \times 5) + (1 \times -3) + 2 \\ &= 0 + 5 - 3 + 2 = 4 \end{aligned}$$

1

Final Result

Because 4 is greater than 0, the activation function
outputs 1



Go to the Party!

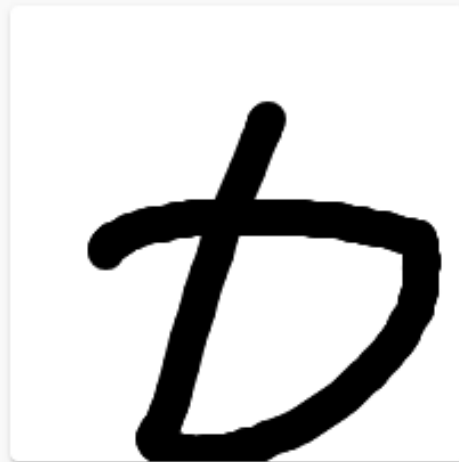
Demo

Digits Recognition (MLP)

Hand-written digits recognition using Multilayer Perceptron (MLP)

 launch  binder  Open in Colab  Stars 1.8k

Draw **one BIG** digit here



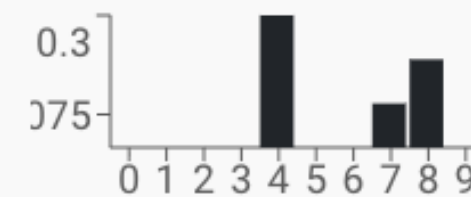
► RECOGNIZE

 CLEAR

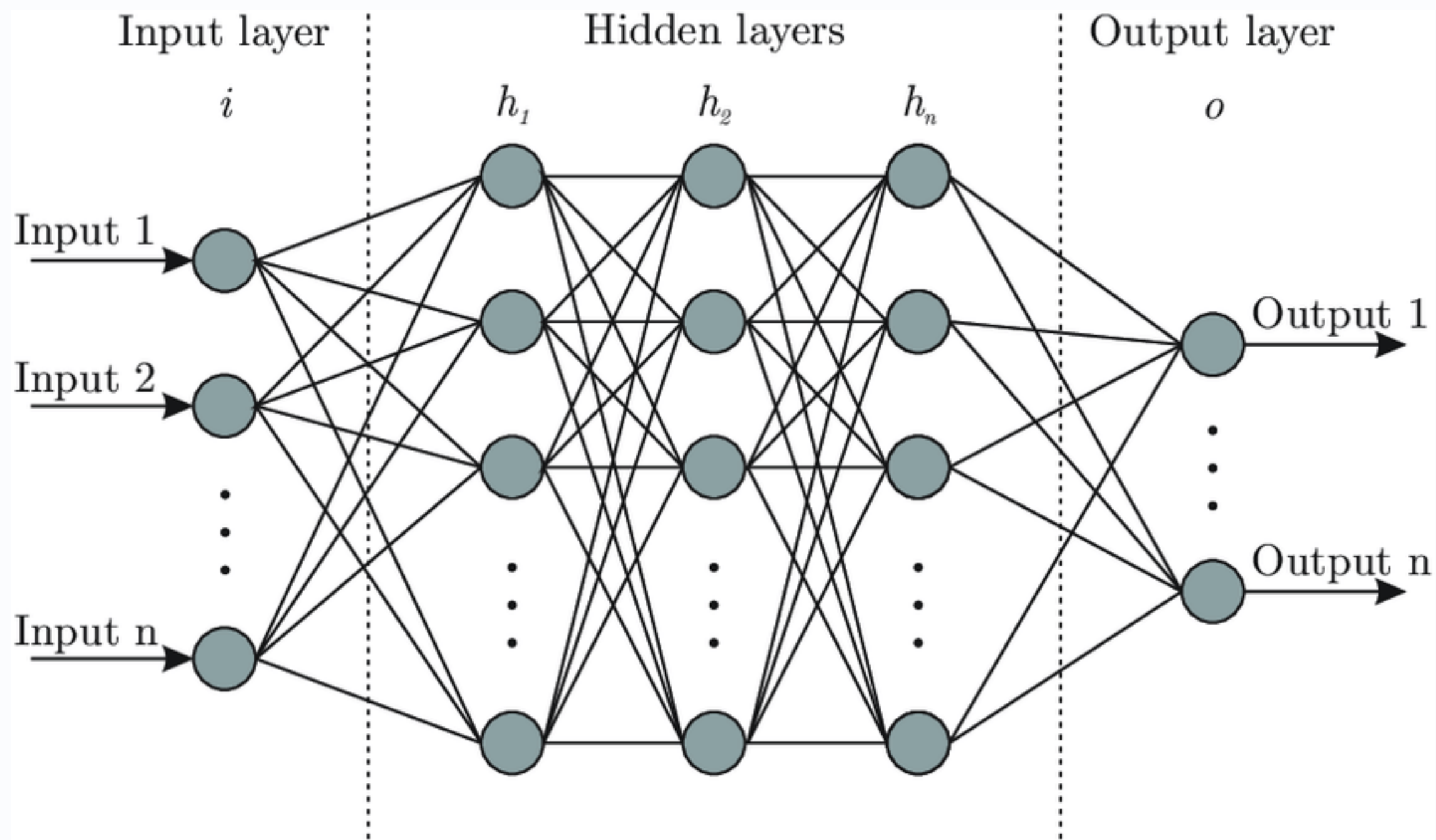
Recognized digit appears here



Probabilities



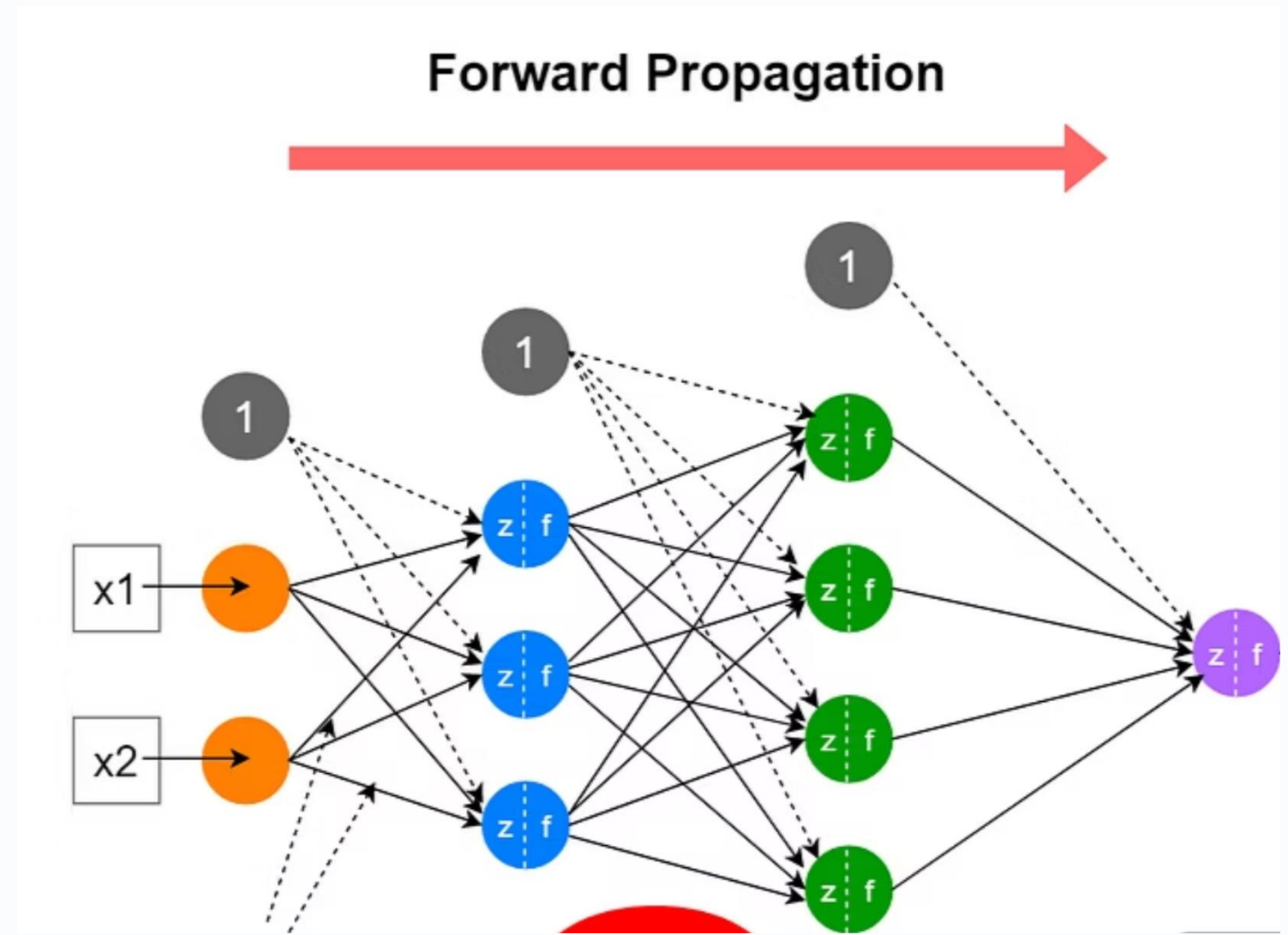
<https://trekhleb.dev/machine-learning-experiments/#/experiments/SketchRecognitionMLP>



How Deep Neural Networks Work

Forward Propagation

The progression of computations through multiple layers of interconnected nodes, each building on the previous layer to refine and optimize the prediction or categorization.

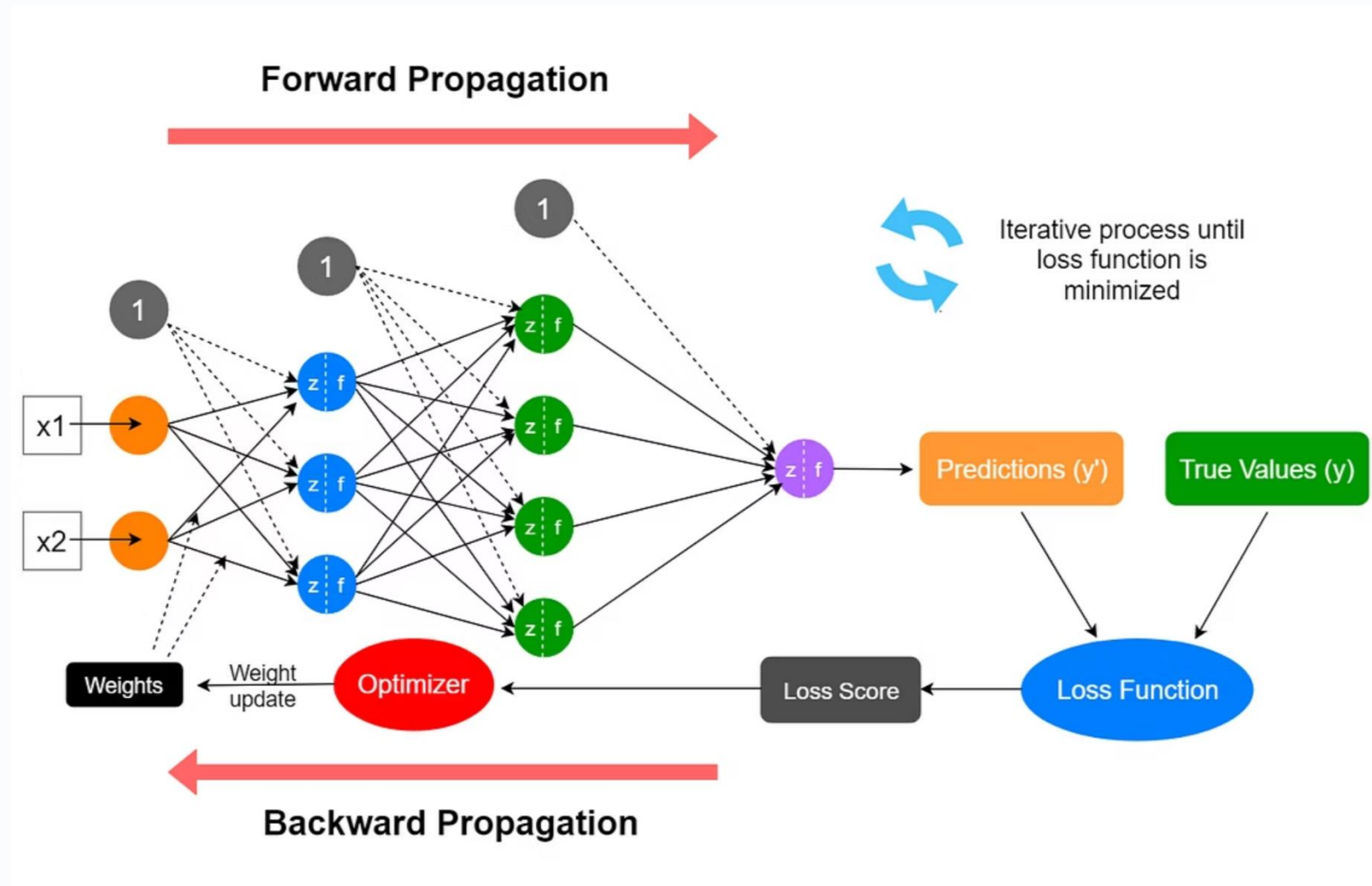


How Deep Neural Networks Work

Backpropagation

Uses algorithms (e.g., gradient descent) to:

- Calculate errors in predictions
- Adjust weights and biases by moving backwards through layers



Together, forward propagation and backpropagation enable a neural network to make predictions and correct for any errors.

Algorithm Examples

There are different types of neural networks to address specific problems or datasets.



CNN

Convolutional Neural Networks

Use several distinct layers that filter different parts of an image before putting it back together

For image processing



RNN

Recurrent Neural Networks

Have “memory” of what happened in the previous layer as contingent to the output of the current layer

For sequential data (natural language processing and speech recognition)

Example: Spam Detection



Early Layers

Identify the individual signal importance of each word or pattern

("prize," "money," "dear,"..)



Middle Layers

Combine the earlier signals to recognize more complex, higher-level cues

(phrases commonly found in spam or unusual sender patterns)



Deeper Layers

Takes all this processed information and computes a single value representing the spam probability of the email

(Either legitimate or spam)

Example: Cat Image Recognition



Early Layers

Detect simple features like edges and colors



Middle Layers

Combine features to recognize shapes like ears, eyes, whiskers



Deeper Layers

Learn full patterns that identify cats

Each layer passes its results to the next, building up understanding step by step.

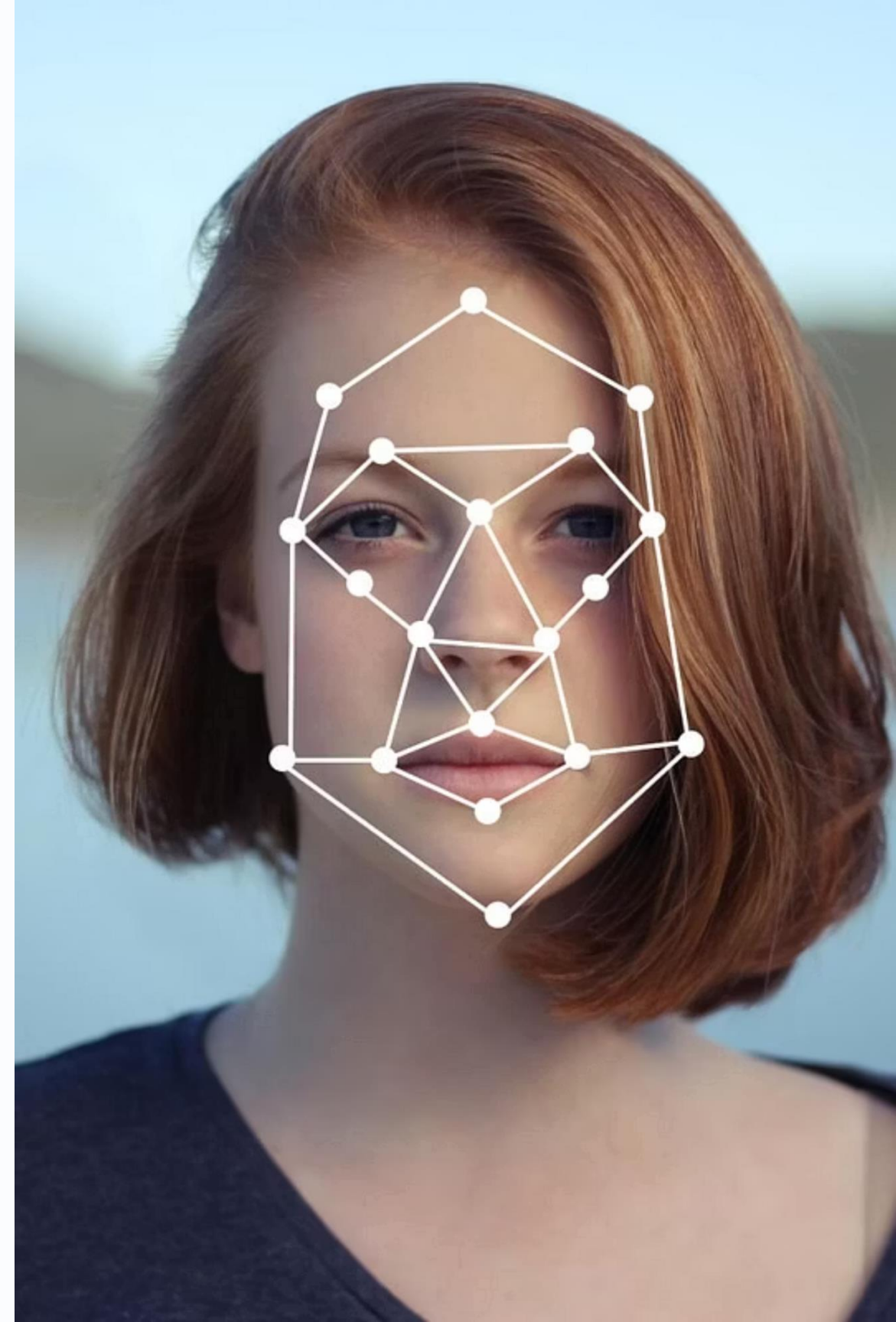
Deep Learning Applications

Face Recognition

The face is made of thousands of pixels.

Too hard to manually tell the computer what features to look at

Deep learning learns all of that by itself



Deep Learning Applications

Language Translation

The meaning of words depend on context

For example, "bank" can mean a financial bank or a river bank

Deep learning can understand the whole sentence to guess the right meaning



Deep Learning Benefits

Learns Features Automatically: No need to manually tell the computer what to look for

Great for Unstructural Data: Works well with images, videos, speech, and text

Works at Scale: Improves with more data



Deep Learning Limitations

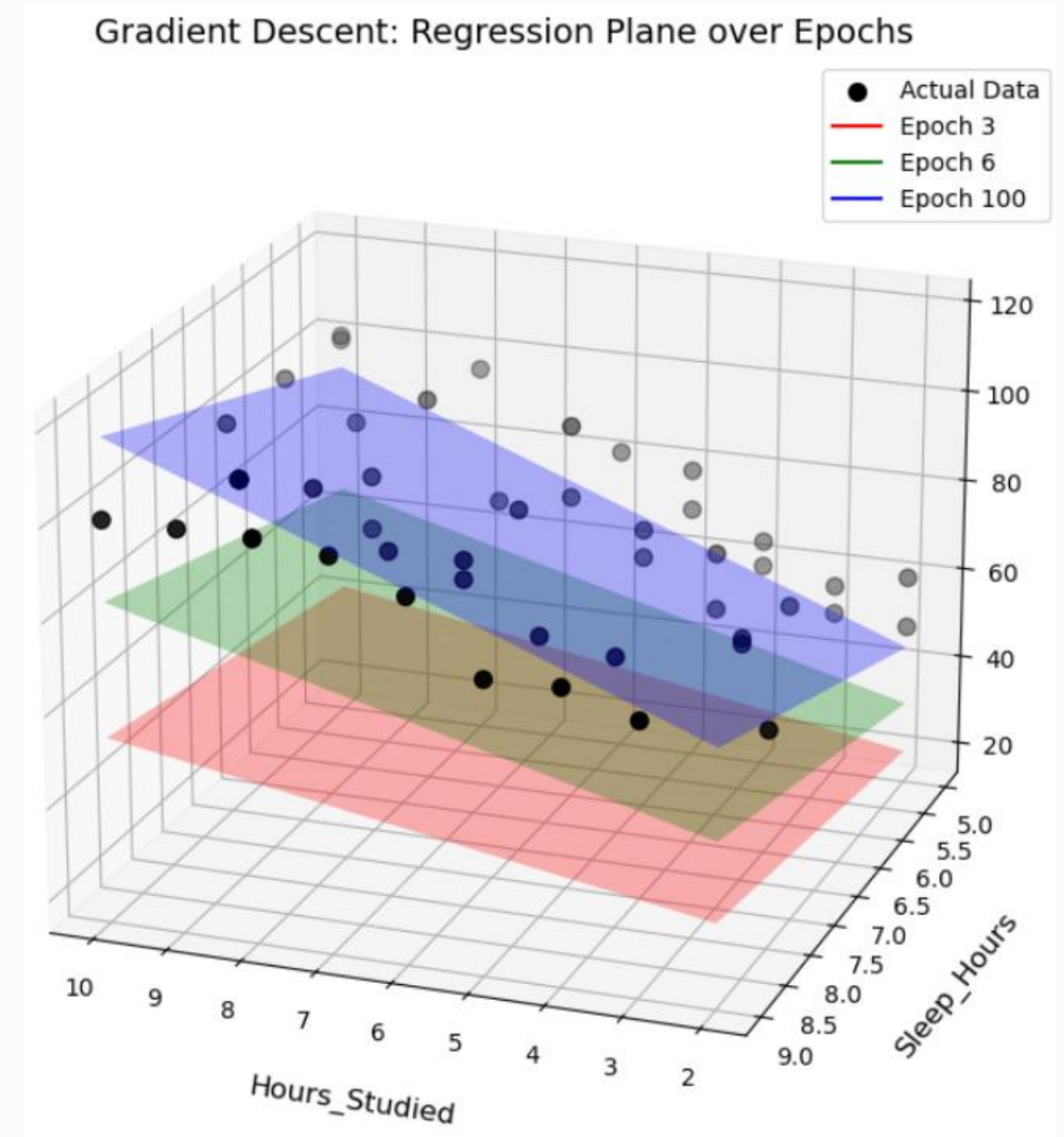
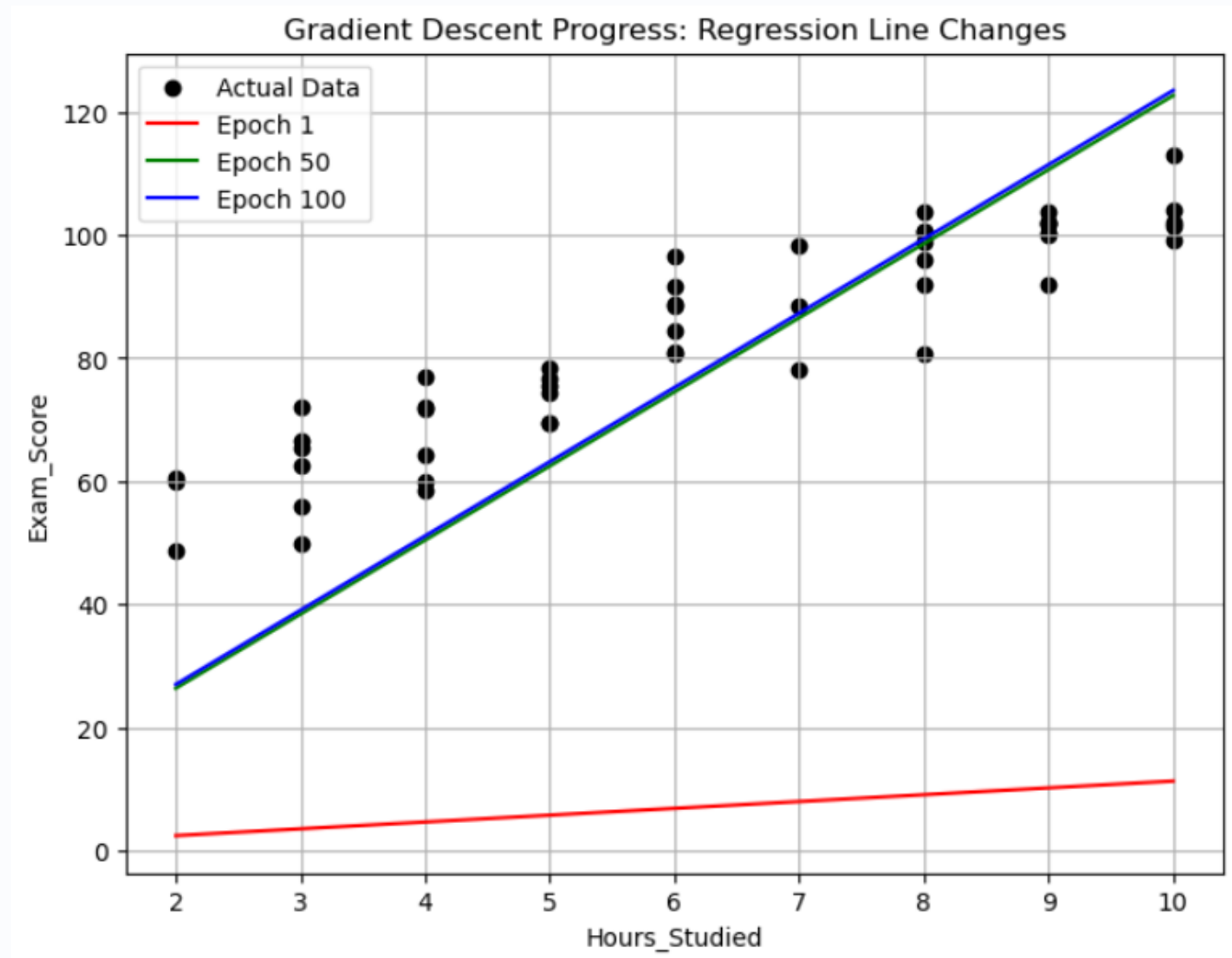
Needs a Lot of Data: Usually requires thousands or millions of examples

Hard to Understand (Black Box): Often unclear how or why the model made a specific decision

Can Overfit: May learn the noise in the data rather than the underlying relationships

Potential Bias: Models can potentially be biased, leading to unfair or inaccurate predictions

ML Implementation Demos



Midterm Exam

Time & Place: In your class time, at your class venue

Bring your laptop: No internet searches; No LLMs (ChatGPT); No WeChat; **Only login to BB**

Cheat paper: One A4 cheat paper (both sides)

Question set: Random questions! **Mind your own question set!**

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Lecture	Date	Content	Remarks
1	2025/9/1-7	Course Overview	
2	2025/9/8-14	Information Technology Developments	
3	2025/9/15-21	AI and Technology Convergence I	
4	2025/9/22-28	AI and Technology Convergence II	
5	2025/9/29-10/12	Machine Learning Essentials I	Two weeks, including National Day Holiday
6	2025/10/13-19	Machine Learning Essentials II	
7	2025/10/20-26	Midterm Assessment	
8	2025/10/27-11/2	Generative AI and Creative Applications	

References

- Machine Learning Crash Course: <https://developers.google.com/machine-learning/crash-course>
- IBM Neural Networks: <https://www.ibm.com/think/topics/neural-networks>
- IBM Reinforcement Learning: <https://www.ibm.com/think/topics/reinforcement-learning>
- Research Articles:
<https://www.nature.com/articles/nature14539>
<https://link.springer.com/article/10.1007/s42979-021-00815-1>
- Demo: <https://developers.google.com/machine-learning/crash-course/neural-networks/interactive-exercises>

Questions?

Course Feedback

